Assignment Day – 5

1. **List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.**

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Component	Description
Motherboard	The main circuit board that connects and allows communication between all computer components
CPU(Processor)	The brain of the computer that performs calculation and executes instructions
CPU Cooler Fan	Keeps the CPU cool by removing heat and preventing overheating
RAM(Random Access Memory)	Temporary memory that stored data and program currently In use for fast access
Storage Drive (HDD,SSD, NVMe SSD)	Stores the operating system, software, and user files permanently, even when the PC is turned off
SMPS (Power Supply Unit)	Convert AC power from the wall into DC power required by the computer components
Graphics Card (GPU)	Processes graphics, videos, and games. Some computer use integrated graphics instead of a dedicated GPU.
Expansion Cards	Additional cards such as Wi-Fi, sound that add extra features to the computer
Cooling Fan	Improve airflow inside the cabinet and help keep all components at safe temperature

Main Components

- **Motherboard**
- **CPU (Processor)**
- **CPU Cooler**

- **RAM**
- **Storage Drive (HDD/SSD/NVMe SSD)**
- **SMPS (Power Supply Unit)**
- **Graphics Card (GPU)**
- **Expansion Card**
- **Cooling Fan**
- **CMOS Battery**

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.
   Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

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1. Unplug Power Cable

Caption: Disconnecting the Power Cable
Explanation: The power cable was unplugged before starting the work.

2. Remove Side Panel

Caption: Removing the Side Panel
Explanation: The side panel was removed to open the desktop case.

3. View Internal Components

Caption: Inspecting Components
Explanation: The internal parts were checked before disassembly.

4. Remove RAM

Caption: Removing the RAM
Explanation: The RAM module was carefully removed from the slot.

5. Remove HDD/SSD

Caption: Removing the Storage Drive
Explanation: The HDD/SSD was disconnected and removed safely.

6. Disconnect Motherboard Cables

Caption: Disconnecting Cables
Explanation: All motherboard cables were unplugged carefully.

7. Remove Motherboard

Caption: Removing the Motherboard
Explanation: The motherboard was unscrewed and removed.

8. Arrange Components

Caption: Organizing Components
Explanation: All components were placed safely for reassembly.

3. **Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.**

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1. **Install the Motherboard**
   - **Explanation: The motherboard was placed in the case and secured with screws.**
2. **Connect Motherboard Cables**
   - **Explanation: The power and front panel cables were connected to the motherboard.**
3. **Install HDD/SSD**
   - **Explanation: The storage drive was installed and connected with SATA cables.**
4. **Install RAM**
   - **Explanation: The RAM module was inserted into the correct memory slot.**
5. **Connect Power Supply Cables**
   - **Explanation: The CPU and 24-pin power cables were connected.**
6. **Check All Connections**
   - **Explanation: All cables and components were checked before closing the case.**
7. **Close the Side Panel**
   - **Explanation: The side panel was attached and tightened with screws.**
8. **Connect Power and Test**
   - **Explanation: The power cable was connected, and the PC was turned on to verify it worked correctly.**

- **Order matters because: The motherboard must be installed first, followed by cables, storage, and RAM. After checking all connections, the case is closed and the PC is tested to ensure everything works properly.**

4. **Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.**

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❖ Safety Precautions

1. **Turn off the PC** – Prevents electrical accidents while working.
2. **Unplug the power cable** – Ensures no power is supplied to the computer.
3. **Press the power button after unplugging** – Removes any remaining electrical charge.
4. **Use an anti-static wrist strap** – Protects components from static electricity.
5. **Work on a clean, dry surface** – Keeps components safe from dust and moisture.
6. **Use a Phillips screwdriver** – Required for removing and tightening screws.
7. **Keep screws in a small container** – Prevents screws from getting lost.
8. **Handle components by the edges** – Avoids damage to electronic parts.
9. **Disconnect cables carefully** – Prevents connectors and wires from breaking.
10. **Check all connections before powering on** – Ensures the PC is assembled correctly and works safely.

❖ Tools Required

1. **Phillips Screwdriver – Used to remove and tighten screws.**
2. **Anti-static Wrist Strap – Prevents static electricity from damaging components.**
3. **Small Container or Magnetic Tray – Keeps screws safe and organized.**
4. **Soft Brush – Removes dust from inside the PC.**
5. **Compressed Air Can – Cleans dust from hard-to-reach areas.**
6. **Cable Ties – Organizes cables neatly during reassembly.**
7. **Flashlight (Optional) – Helps see small connectors inside the case.**